

Physics

This section presents the requirements for programs in:

- **Physics (Astrophysics Stream) B.Sc. Honours**
- **Physics (Experimental Physics Stream) B.Sc. Honours**
- **Physics (Theory Stream) B.Sc. Honours**
- **Physics B.Sc. Major**
- **Physics B.Sc.**
- **Applied Physics B.Sc. Honours**
- **Mathematics and Physics B.Sc. Double Honours**
- **Biology and Physics B.Sc. Combined Honours**
- **Chemistry and Physics B.Sc. Combined Honours**
- **Minor in Physics**

The Department of Physics also offers the program: Engineering Physics - B.Eng. Consult the Engineering program section for details about this program.

Program Requirements

Course Categories for Physics

The program descriptions below make use of the following course categories, which are defined in the B.Sc. Regulations section.

- Approved Courses Outside the Faculties of Science and Engineering and Design
- Free Elective

Physics (Astrophysics Stream)

B.Sc. Honours (20.0 credits)

A. Credits Included in the Major CGPA (11.0 credits)

1. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I & PHYS 1002 [0.5] Foundations of Physics II	
PHYS 1003 [0.5] Introductory Mechanics and & PHYS 1004 [0.5] Thermodynamics Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II (with an average grade of B- or higher)	
2. 3.0 credits in:	3.0
PHYS 2007 [0.5] Second Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 2203 [0.5] Astronomy	
PHYS 2212 [0.5] Wave Mechanics and Thermodynamics	
PHYS 2305 [0.5] Electricity and Magnetism	
PHYS 2605 [0.5] Modern Physics I	
PHYS 2801 [0.5] Computational Methods in Physics	
3. 5.0 credits in:	5.0
PHYS 3009 [0.5] Third Year Physics Laboratory: Selected Experiments and Seminars with Observational Astronomy	
PHYS 3308 [0.5] Electromagnetism	

PHYS 3606 [0.5] Modern Physics II	
PHYS 3701 [0.5] Elements of Quantum Mechanics	
PHYS 3802 [0.5] Advanced Dynamics	
PHYS 3807 [0.5] Mathematical Physics I	
PHYS 4201 [0.5] Astrophysics	
PHYS 4202 [0.5] Cosmology	
PHYS 4409 [0.5] Thermodynamics and Statistical Physics	
PHYS 4707 [0.5] Introduction to Quantum Mechanics I	
4. 1.0 credit from:	1.0
a. PHYS 4907 plus 0.5 credit 4000-level PHYS	
b. PHYS 4908 plus 0.5 credit 4000-level PHYS	
c. PHYS 4909 [1.0]	
5. 0.5 credit in PHYS at the 4000-level or above	0.5
6. 0.5 credit in PHYS, COMP, MATH and/or STAT at the 3000-level or above	0.5
B. Credits Not Included in the Major CGPA (9.0 credits)	
7. 1.0 credit from:	1.0
BIOL 1103 [0.5] Foundations of Biology I & BIOL 1104 [0.5] Foundations of Biology II	
CHEM 1001 [0.5] General Chemistry I & CHEM 1002 [0.5] General Chemistry II	
ERTH 1002 [0.5] The Earth and Life Odyssey: A & ERTH 2312 [0.5] Journey Through Billions of Years Paleontology	
8. 3.5 credits in:	3.5
MATH 1004 [0.5] Calculus for Engineering or Physics	
MATH 1005 [0.5] Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5] Linear Algebra for Engineering or Science	
MATH 2004 [0.5] Multivariable Calculus for Engineering or Physics	
MATH 2107 [0.5] Linear Algebra II	
MATH 3705 [0.5] Mathematical Methods I	
STAT 3502 [0.5] Probability and Statistics	
9. 0.5 credit in:	0.5
MATH 3800 [0.5] Mathematical Modeling and Computational Methods	
10. 1.0 credit in:	1.0
COMP 1005 [0.5] Introduction to Computer Science I	
COMP 1006 [0.5] Introduction to Computer Science II	
11. 0.5 credit in:	0.5
ISAP 1000 [0.5] Seminar in Science	
Approved courses outside the faculties of Science and Engineering and Design	
12. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design	1.5
13. 1.0 credit in free electives	1.0
Total Credits	20.0

Physics (Experimental Physics Stream)

B.Sc. Honours (20.0 credits)

A. Credits Included in the Major CGPA (11.5 credits)

1. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I & PHYS 1002 [0.5] Foundations of Physics II (recommended)	

PHYS 1003 [0.5] & PHYS 1004 [0.5]	Introductory Mechanics and Thermodynamics Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] & PHYS 1008 [0.5]	Elementary University Physics I Elementary University Physics II (with an average grade of B- or higher)	
2. 2.5 credits in:		2.5
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 2212 [0.5]	Wave Mechanics and Thermodynamics	
PHYS 2305 [0.5]	Electricity and Magnetism	
PHYS 2605 [0.5]	Modern Physics I	
PHYS 2801 [0.5]	Computational Methods in Physics	
3. 1.0 credit in:		1.0
ELEC 2501 [0.5]	Circuits and Signals	
ELEC 2507 [0.5]	Electronics I	
4. 4.5 credits in:		4.5
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 3308 [0.5]	Electromagnetism	
PHYS 3606 [0.5]	Modern Physics II	
PHYS 3701 [0.5]	Elements of Quantum Mechanics	
PHYS 3802 [0.5]	Advanced Dynamics	
PHYS 3807 [0.5]	Mathematical Physics I	
PHYS 4008 [0.5]	Fourth-Year Physics Laboratory: Selected Experiments and Workshop	
PHYS 4409 [0.5]	Thermodynamics and Statistical Physics	
PHYS 4707 [0.5]	Introduction to Quantum Mechanics I	
5. 1.0 credit from:		1.0
a. PHYS 4907 [0.5] plus 0.5 credit 4000-level PHYS		
b. PHYS 4908 [0.5] plus 0.5 credit 4000-level PHYS		
c. PHYS 4909 [1.0]		
6. 1.0 credit in 4000-level or above PHYS (PHYS 4807 is recommended for 0.5 credit)		1.0
7. 0.5 credit in 3000-level or above PHYS, COMP, ELEC, MATH and/or STAT		0.5
B. Credits Not Included In the Major CGPA (8.5 credits)		
8. 1.0 credit from:		1.0
BIOL 1103 [0.5] & BIOL 1104 [0.5]	Foundations of Biology I Foundations of Biology II	
CHEM 1001 [0.5] & CHEM 1002 [0.5]	General Chemistry I General Chemistry II	
ERTH 1002 [0.5] & ERTH 2312 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years Paleontology	
9. 3.0 credits in:		3.0
MATH 1004 [0.5]	Calculus for Engineering or Physics	
MATH 1005 [0.5]	Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5]	Linear Algebra for Engineering or Science	

MATH 2004 [0.5]	Multivariable Calculus for Engineering or Physics	
MATH 3705 [0.5]	Mathematical Methods I	
STAT 3502 [0.5]	Probability and Statistics	
10. 0.5 credit in:		0.5
MATH 3800 [0.5]	Mathematical Modeling and Computational Methods	
11. 1.0 credit in:		1.0
COMP 1005 [0.5]	Introduction to Computer Science I	
COMP 1006 [0.5]	Introduction to Computer Science II	
12. 0.5 credit from:		0.5
ISAP 1000 [0.5]	Seminar in Science	
Approved courses outside the faculties of Science and Engineering and Design		
13. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design		1.5
14. 1.0 credit in free electives		1.0
Total Credits		20.0

Physics (Theory Stream) B.Sc. Honours (20.0 credits)

A. Credits Included in the Major CGPA (11.0 credits)		
1. 1.0 credit from:		1.0
PHYS 1001 [0.5] & PHYS 1002 [0.5]	Foundations of Physics I Foundations of Physics II (recommended)	
PHYS 1003 [0.5] & PHYS 1004 [0.5]	Introductory Mechanics and Thermodynamics Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] & PHYS 1008 [0.5]	Elementary University Physics I Elementary University Physics II (with an average grade of B- or higher)	
2. 2.5 credits in:		2.5
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 2212 [0.5]	Wave Mechanics and Thermodynamics	
PHYS 2305 [0.5]	Electricity and Magnetism	
PHYS 2605 [0.5]	Modern Physics I	
PHYS 2801 [0.5]	Computational Methods in Physics	
3. 4.5 credits in:		4.5
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 3308 [0.5]	Electromagnetism	
PHYS 3606 [0.5]	Modern Physics II	
PHYS 3701 [0.5]	Elements of Quantum Mechanics	
PHYS 3802 [0.5]	Advanced Dynamics	
PHYS 3807 [0.5]	Mathematical Physics I	
PHYS 4409 [0.5]	Thermodynamics and Statistical Physics	
PHYS 4707 [0.5]	Introduction to Quantum Mechanics I	
PHYS 4708 [0.5]	Introduction to Quantum Mechanics II	
4. 1.0 credit from:		1.0
a. PHYS 4907 plus 0.5 credit 4000-level PHYS		

b. PHYS 4908 plus 0.5 credit 4000-level PHYS	
c. PHYS 4909 [1.0]	
5. 1.0 credit in PHYS at the 4000-level or above	1.0
6. 1.0 credit in PHYS, COMP, MATH and/or STAT at the 3000-level or above	1.0
B. Credits Not Included In the Major CGPA (9.0 credits)	
7. 1.0 credit from:	1.0
BIOL 1103 [0.5] Foundations of Biology I & BIOL 1104 [0.5] Foundations of Biology II	
CHEM 1001 [0.5] General Chemistry I & CHEM 1002 [0.5] General Chemistry II	
ERTH 1002 [0.5] The Earth and Life Odyssey: A & ERTH 2312 [0.5] Journey Through Billions of Years Paleontology	
8. 3.5 credits in:	3.5
MATH 1004 [0.5] Calculus for Engineering or Physics	
MATH 1005 [0.5] Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5] Linear Algebra for Engineering or Science	
MATH 2004 [0.5] Multivariable Calculus for Engineering or Physics	
MATH 2107 [0.5] Linear Algebra II	
MATH 3705 [0.5] Mathematical Methods I	
STAT 3502 [0.5] Probability and Statistics	
9. 0.5 credit in:	0.5
MATH 3800 [0.5] Mathematical Modeling and Computational Methods	
10. 1.0 credit in:	1.0
COMP 1005 [0.5] Introduction to Computer Science I	
COMP 1006 [0.5] Introduction to Computer Science II	
11. 0.5 credit in:	0.5
ISAP 1000 [0.5] Seminar in Science	
or approved courses outside the faculties of Science and Engineering and Design	
12. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design	1.5
13. 1.0 credit in free electives	1.0
Total Credits	20.0

Physics

B.Sc. Major (20.0 credits)

A. Credits Included in the Major CGPA (9.0 credits)

1. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I & PHYS 1002 [0.5] Foundations of Physics II (recommended)	
PHYS 1003 [0.5] Introductory Mechanics and & PHYS 1004 [0.5] Thermodynamics Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II (with an average grade of B- or higher)	
2. 2.5 credits in:	2.5
PHYS 2007 [0.5] Second Year Physics Laboratory: Selected Experiments and Seminars	

PHYS 2212 [0.5] Wave Mechanics and Thermodynamics	
PHYS 2305 [0.5] Electricity and Magnetism	
PHYS 2605 [0.5] Modern Physics I	
PHYS 2801 [0.5] Computational Methods in Physics	
3. 0.5 credit in approved computer science, engineering, mathematics or statistics electives at the 2000-level or higher which may include 1000-level computer science	0.5
4. 2.0 credits in:	2.0
PHYS 3007 [0.5] Third Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 3308 [0.5] Electromagnetism	
PHYS 3606 [0.5] Modern Physics II or PHYS 3608 [0] Modern Applied Physics	
PHYS 3701 [0.5] Elements of Quantum Mechanics	
5. 1.0 credit in PHYS at the 4000-level	1.0
6. 1.5 credit in PHYS at the 3000-level or above	1.5
7. 0.5 credit in ELEC and/or science faculty electives (excluding TSES) at the 3000-level or above	0.5
B. Credits Not Included In the Major CGPA (11.0 credits)	
8. 1.0 credit from:	1.0
BIOL 1103 [0.5] Foundations of Biology I & BIOL 1104 [0.5] Foundations of Biology II	
CHEM 1001 [0.5] General Chemistry I & CHEM 1002 [0.5] General Chemistry II	
ERTH 1002 [0.5] The Earth and Life Odyssey: A & ERTH 2312 [0.5] Journey Through Billions of Years Paleontology	
9. 3.0 credits in:	3.0
MATH 1004 [0.5] Calculus for Engineering or Physics	
MATH 1005 [0.5] Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5] Linear Algebra for Engineering or Science	
MATH 2004 [0.5] Multivariable Calculus for Engineering or Physics	
MATH 3705 [0.5] Mathematical Methods I	
STAT 2507 [0.5] Introduction to Statistical Modeling I or STAT 3502 [0.5] Probability and Statistics	
10. 0.5 credit in:	0.5
COMP 1005 [0.5] Introduction to Computer Science I	
11. 3.5 credits in Advanced Science Faculty Electives and/or approved courses outside the Faculties of Science and Engineering selected in consultation with the Department to complement the study of physics; these credits may be used with an additional 0.5 credit to complete the requirements of a minor designation	3.5
12. 0.5 credit from:	0.5
ISAP 1000 [0.5] Seminar in Science	
Approved courses outside the faculties of Science and Engineering and Design	
13. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design	1.5
14. 1.0 credit in free electives	1.0
Total Credits	20.0

Physics

B.Sc. (15.0 credits)

A. Credits Included in the Major CGPA (6.5 credits)

1. 1.0 credit from: 1.0

PHYS 1001 [0.5] Foundations of Physics I
& PHYS 1002 [0.5] Foundations of Physics II (recommended)

PHYS 1003 [0.5] Introductory Mechanics and
& PHYS 1004 [0.5] Thermodynamics
Introductory Electromagnetism and
Wave Motion

PHYS 1007 [0.5] Elementary University Physics I
& PHYS 1008 [0.5] Elementary University Physics II
(with an average grade of B- or
higher)

2. 4.0 credits in: 4.0

PHYS 2007 [0.5] Second Year Physics Laboratory:
Selected Experiments and
Seminars

PHYS 2212 [0.5] Wave Mechanics and
Thermodynamics

PHYS 2305 [0.5] Electricity and Magnetism

PHYS 2605 [0.5] Modern Physics I

PHYS 2801 [0.5] Computational Methods in Physics

PHYS 3308 [0.5] Electromagnetism

PHYS 3701 [0.5] Elements of Quantum Mechanics

PHYS 3802 [0.5] Advanced Dynamics

3. 0.5 credit from: 0.5

PHYS 3007 [0.5] Third Year Physics Laboratory:
Selected Experiments and
Seminars

PHYS 3606 [0.5] Modern Physics II
or PHYS 3608 [0.5] Modern Applied Physics

4. 1.0 credit in PHYS at the 3000-level or above 1.0

B. Credits Not Included in the Major CGPA (8.5 credits)

5. 2.5 credits in: 2.5

MATH 1004 [0.5] Calculus for Engineering or Physics

MATH 1005 [0.5] Differential Equations and Infinite
Series for Engineering or Physics

MATH 1104 [0.5] Linear Algebra for Engineering or
Science

MATH 2004 [0.5] Multivariable Calculus for
Engineering or Physics

MATH 3705 [0.5] Mathematical Methods I

6. 1.0 credit from: 1.0

BIOL 1103 [0.5] Foundations of Biology I
& BIOL 1104 [0.5] Foundations of Biology II

CHEM 1001 [0.5] General Chemistry I
&
CHEM 1002 [0.5] General Chemistry II

ERTH 1002 [0.5] The Earth and Life Odyssey: A
& ERTH 2312 [0.5] Journey Through Billions of Years
Paleontology

7. 0.5 credit in: 0.5

COMP 1005 [0.5] Introduction to Computer Science I

8. 1.0 credit in Science Continuation Courses (not PHYS) 1.0

9. 0.5 credit in Science Faculty Electives and/or Science Continuation Courses 0.5

10. 2.0 credits in ISAP 1000 or approved courses outside the faculties of Science and Engineering and Design 2.0

11. 1.0 credit in free electives 1.0

Total Credits 15.0

Applied Physics

B.Sc. Honours (20.0 credits)

A. Credits Included in the Major CGPA (11.5 credits)

1. 1.0 credit from: 1.0

PHYS 1001 [0.5] Foundations of Physics I
& PHYS 1002 [0.5] Foundations of Physics II
(recommended)

PHYS 1003 [0.5] Introductory Mechanics and
& PHYS 1004 [0.5] Thermodynamics
Introductory Electromagnetism and
Wave Motion

PHYS 1007 [0.5] Elementary University Physics I
& PHYS 1008 [0.5] Elementary University Physics II
(with an average grade of B- or
higher)

2. 2.5 credits in: 2.5

PHYS 2007 [0.5] Second Year Physics Laboratory:
Selected Experiments and
Seminars

PHYS 2212 [0.5] Wave Mechanics and
Thermodynamics

PHYS 2305 [0.5] Electricity and Magnetism

PHYS 2605 [0.5] Modern Physics I

PHYS 2801 [0.5] Computational Methods in Physics

3. 1.0 credit in: 1.0

ELEC 2501 [0.5] Circuits and Signals

ELEC 2507 [0.5] Electronics I

4. 0.5 credit from: 0.5

ECOR 2606 [0.5] Numerical Methods

MATH 3800 [0.5] Mathematical Modeling and
Computational Methods

5. 4.0 credits in: 4.0

PHYS 3007 [0.5] Third Year Physics Laboratory:
Selected Experiments and
Seminars

PHYS 3308 [0.5] Electromagnetism

PHYS 3608 [0.5] Modern Applied Physics

PHYS 3701 [0.5] Elements of Quantum Mechanics

PHYS 3802 [0.5] Advanced Dynamics

PHYS 3807 [0.5] Mathematical Physics I

PHYS 4008 [0.5] Fourth-Year Physics Laboratory:
Selected Experiments and
Workshop

PHYS 4707 [0.5] Introduction to Quantum Mechanics
I

6. 1.0 credit from: 1.0

PHYS 3207 [0.5] Topics in Biophysics

PHYS 4203 [0.5] Physical Applications of Fourier
Analysis

PHYS 4208 [0.5] Modern Optics

PHYS 4608 [0.5] Nuclear Physics

PHYS 4807 [0.5] Statistical Data Analysis
Techniques for Physics

7. 0.5 credit from: 0.5

ELEC 3509 [0.5] Electronics II

ELEC 3908 [0.5] Physical Electronics

COMP at the 3000-level

PHYS at the 4000-level	
8. 1.0 credit from:	1.0
a. PHYS 4907 plus 0.5 credit 4000-level PHYS	
b. PHYS 4908 plus 0.5 credit 4000-level PHYS	
c. PHYS 4909 [1.0]	
B. Credits Not Included in the Major CGPA (8.5 credits)	
9. 1.0 credit from:	1.0
BIOL 1103 [0.5] Foundations of Biology I	
& BIOL 1104 [0.5] Foundations of Biology II	
CHEM 1001 [0.5] General Chemistry I	
& CHEM 1002 [0.5] General Chemistry II	
ERTH 1002 [0.5] The Earth and Life Odyssey: A	
& ERTH 2312 [0.5] Journey Through Billions of Years	
Paleontology	
10. 3.0 credits in:	3.0
MATH 1004 [0.5] Calculus for Engineering or Physics	
MATH 1005 [0.5] Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5] Linear Algebra for Engineering or Science	
MATH 2004 [0.5] Multivariable Calculus for Engineering or Physics	
STAT 3502 [0.5] Probability and Statistics	
MATH 3705 [0.5] Mathematical Methods I	
11. 0.5 credit in:	0.5
COMP 1005 [0.5] Introduction to Computer Science I	
12. 3.5 credits in:	3.5
a. (COMP 1006 and COMP 2401) or (SYSC 2006 and SYSC 2004)	
b. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design	
c. 1.0 credit in free electives	
13. 0.5 credit from:	0.5
ISAP 1000 [0.5] Seminar in Science	
Approved courses outside the faculties of Science and Engineering and Design	
Total Credits	20.0

Mathematics and Physics

B.Sc. Double Honours (21.5 credits)

Note that the following courses have minimum grade requirements in their prerequisites. Refer to the section Course Prerequisites under the Mathematics and Statistics programs sections of the calendar.

MATH 2001 [0.5]	Multivariable Calculus and Fundamentals of Analysis I
MATH 2002 [0.5]	Multivariable Calculus and Fundamentals of Analysis II
MATH 2100 [1.0]	Algebra
MATH 2454 [0.5]	Ordinary Differential Equations (Honours)
STAT 2655 [0.5]	Introduction to Probability with Applications (Honours)

A. Credits Included in the Major CGPA (17.0 credits)

1. 7.5 credits in:	7.5
MATH 1052 [0.5]	Calculus and Introductory Analysis I
MATH 1152 [0.5]	Introductory Algebra I

MATH 1800 [0.5]	Introduction to Mathematical Reasoning
MATH 2001 [0.5]	Multivariable Calculus and Fundamentals of Analysis I
MATH 2002 [0.5]	Multivariable Calculus and Fundamentals of Analysis II
MATH 2052 [0.5]	Calculus and Introductory Analysis II
MATH 2100 [1.0]	Algebra
MATH 2152 [0.5]	Introductory Algebra II
MATH 2454 [0.5]	Ordinary Differential Equations (Honours)
MATH 3001 [0.5]	Real Analysis I (Honours)
MATH 3057 [0.5]	Functions of a Complex Variable (Honours)
MATH 3702 [0.5]	Mathematical Modelling (Honours)
MATH 3705 [0.5]	Mathematical Methods I
STAT 2655 [0.5]	Introduction to Probability with Applications (Honours)
2. 0.5 credit from:	0.5
MATH 3002 [0.5]	Real Analysis II (Honours)
MATH 3003 [0.5]	Advanced Differential Calculus (Honours)
MATH 3106 [0.5]	Introduction to Group Theory (Honours)
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3606 [0.5]	Modern Physics II
3. 1.0 credit in 4000-level or higher MATH, STAT	1.0
4. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I	
& PHYS 1002 [0.5] Foundations of Physics II (recommended)	
PHYS 1003 [0.5] Introductory Mechanics and	
& PHYS 1004 [0.5] Thermodynamics	
Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] Elementary University Physics I	
& PHYS 1008 [0.5] Elementary University Physics II (with an average grade of B- or higher)	
5. 2.0 credits in:	2.0
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars
PHYS 2212 [0.5]	Wave Mechanics and Thermodynamics
PHYS 2305 [0.5]	Electricity and Magnetism
PHYS 2605 [0.5]	Modern Physics I
6. 3.0 credits in:	3.0
PHYS 3308 [0.5]	Electromagnetism
PHYS 3701 [0.5]	Elements of Quantum Mechanics
PHYS 3802 [0.5]	Advanced Dynamics
PHYS 4409 [0.5]	Thermodynamics and Statistical Physics
PHYS 4707 [0.5]	Introduction to Quantum Mechanics I
PHYS 4708 [0.5]	Introduction to Quantum Mechanics II
7. 1.0 credit in PHYS at the 4000-level	1.0

8. 1.0 credit from:	1.0
a. MATH 4905 or PHYS 4907 or PHYS 4908 plus 0.5 credit 4000-level MATH or PHYS	
b. PHYS 4909 [1.0]	

B. Credits Not Included in the Major CGPA (4.5 credits)

9. 1.0 credit from:	1.0
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BIOL 1103 [0.5]	Foundations of Biology I
& BIOL 1104 [0.5]	Foundations of Biology II
CHEM 1001 [0.5]	General Chemistry I
&	General Chemistry II
CHEM 1002 [0.5]	
ERTH 1002 [0.5]	The Earth and Life Odyssey: A
& ERTH 2312 [0.5]	Journey Through Billions of Years
	Paleontology

10. 0.5 credit in:	0.5
COMP 1005 [0.5]	Introduction to Computer Science I

11. 0.5 credit from:	0.5
ISAP 1000 [0.5]	Seminar in Science
Approved courses outside the faculties of Science and Engineering and Design	

12. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design	1.5
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13. 1.0 credit in free electives	1.0
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Total Credits	21.5
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Biology and Physics

B.Sc. Combined Honours (20.0 credits)

A. Credits Included in the Major CGPA (12.5 credits)

1. 1.0 credit from:	1.0
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PHYS 1001 [0.5]	Foundations of Physics I
& PHYS 1002 [0.5]	Foundations of Physics II (recommended)
PHYS 1003 [0.5]	Introductory Mechanics and
& PHYS 1004 [0.5]	Thermodynamics
	Introductory Electromagnetism and
	Wave Motion
PHYS 1007 [0.5]	Elementary University Physics I
& PHYS 1008 [0.5]	Elementary University Physics II (with an average grade of B- or higher)

2. 3.5 credits in:	3.5
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PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars
PHYS 2212 [0.5]	Wave Mechanics and Thermodynamics
PHYS 2305 [0.5]	Electricity and Magnetism
PHYS 2605 [0.5]	Modern Physics I
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3207 [0.5]	Topics in Biophysics
PHYS 3701 [0.5]	Elements of Quantum Mechanics

3. 1.0 credit from:	1.0
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PHYS 3308 [0.5]	Electromagnetism
PHYS 3606 [0.5]	Modern Physics II
PHYS 3802 [0.5]	Advanced Dynamics

4. 1.0 credit from:	1.0
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PHYS 3807 [0.5]	Mathematical Physics I
PHYS 4203 [0.5]	Physical Applications of Fourier Analysis

PHYS 4409 [0.5]	Thermodynamics and Statistical Physics
PHYS 4707 [0.5]	Introduction to Quantum Mechanics I

5. 4.0 credits from:	4.0
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BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
BIOL 2200 [0.5]	Cellular Biochemistry
BIOL 2104 [0.5]	Introductory Genetics
BIOL 2001 [0.5]	Animals: Form and Function
BIOL 2002 [0.5]	Plants: Form and Function
BIOL 3201 [0.5]	Cell Biology
BIOL 3104 [0.5]	Molecular Genetics
BIOL 3305 [0.5]	Human and Comparative Physiology

6. 1.0 credit from:	1.0
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BIOL 3501 [0.5]	Biomechanics
BIOL 4106 [0.5]	Advances in Molecular Biology
BIOL 4109 [0.5]	Laboratory Techniques in Molecular Genetics

BIOL 4201 [0.5]	Advanced Cell Culture and Tissue Engineering
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BIOL 4202 [0.5]	Mutagenesis and DNA Repair
BIOL 4301 [0.5]	Current Topics in Biotechnology
BIOL 4306 [0.5]	Animal Neurophysiology
BIOL 4309 [0.5]	Studies in Human Performance
BIOL 4319 [0.5]	Studies in Exercise Physiology

7. 1.0 credit from:	1.0
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BIOL 4905 [1.0]	Honours Workshop
BIOL 4907 [1.0]	Honours Essay and Research Proposal
BIOL 4908 [1.0]	Honours Research Thesis
PHYS 4909 [1.0]	Fourth-Year Project
PHYS 4907 plus 0.5 credit 4000-level PHYS	
PHYS 4908 plus 0.5 credit 4000-level PHYS	

B. Credits Not Included in the Major CGPA (7.5 credits)

8. 1.0 credit in:	1.0
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CHEM 1001 [0.5]	General Chemistry I
&	General Chemistry II
CHEM 1002 [0.5]	

9. 1.5 credits in:	1.5
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MATH 1004 [0.5]	Calculus for Engineering or Physics
MATH 1005 [0.5]	Differential Equations and Infinite Series for Engineering or Physics
MATH 1104 [0.5]	Linear Algebra for Engineering or Science

10. 2.0 credits in:	2.0
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STAT 2507 [0.5]	Introduction to Statistical Modeling I
MATH 2004 [0.5]	Multivariable Calculus for Engineering or Physics
MATH 3705 [0.5]	Mathematical Methods I
MATH 3800 [0.5]	Mathematical Modeling and Computational Methods

11. 0.5 credit in:	0.5
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COMP 1005 [0.5]	Introduction to Computer Science I
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12. 2.0 credits in approved courses outside the faculties of Science and Engineering and Design (may include ISAP 1000)	2.0
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13. 0.5 credit in free electives	0.5
Total Credits	20.0
Chemistry and Physics	
B.Sc. Combined Honours (20.0 credits)	
A. Credits Included in the Major CGPA (13.0 credits)	
1. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I & PHYS 1002 [0.5] Foundations of Physics II (recommended)	
PHYS 1003 [0.5] Introductory Mechanics and & PHYS 1004 [0.5] Thermodynamics Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II (with an average grade of B- or higher)	
2. 3.5 credits in:	3.5
PHYS 2007 [0.5] Second Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 2212 [0.5] Wave Mechanics and Thermodynamics	
PHYS 2305 [0.5] Electricity and Magnetism	
PHYS 2605 [0.5] Modern Physics I	
PHYS 3007 [0.5] Third Year Physics Laboratory: Selected Experiments and Seminars	
PHYS 3701 [0.5] Elements of Quantum Mechanics	
PHYS 3807 [0.5] Mathematical Physics I	
3. 1.5 credits from:	1.5
PHYS 3308 [0.5] Electromagnetism	
PHYS 3606 [0.5] Modern Physics II	
PHYS 3802 [0.5] Advanced Dynamics	
PHYS 4707 [0.5] Introduction to Quantum Mechanics I	
4. 4.5 credits in:	4.5
CHEM 1011 [0.5] Enriched General Chemistry 1	
CHEM 1012 [0.5] Enriched General Chemistry 2	
CHEM 2103 [0.5] Physical Chemistry I	
CHEM 2104 [0.5] Physical Chemistry II	
CHEM 2203 [0.5] Organic Chemistry I	
CHEM 2204 [0.5] Organic Chemistry II	
CHEM 2501 [0.5] Introduction to Inorganic and Bioinorganic Chemistry	
CHEM 3102 [0.5] Methods in Computational Chemistry	
CHEM 3503 [0.5] Inorganic Chemistry I	
5. 0.5 credit in:	0.5
CHEM 3107 [0.5] Experimental Methods in Nanoscience	
6. 1.0 credit in CHEM at the 4000-level	1.0
7. 1.0 credit from:	1.0
CHEM 4908 [1.0] Research Project and Seminar	
PHYS 4909 [1.0] Fourth-Year Project	
PHYS 4907 plus 0.5 credit in PHYS at the 4000-level	
PHYS 4908 plus 0.5 credit in PHYS at the 4000-level	
B. Credits Not Included in the Major CGPA (7.0 credits)	
8. 3.0 credits in:	3.0

MATH 1004 [0.5]	Calculus for Engineering or Physics	
MATH 1005 [0.5]	Differential Equations and Infinite Series for Engineering or Physics	
MATH 1104 [0.5]	Linear Algebra for Engineering or Science	
MATH 2004 [0.5]	Multivariable Calculus for Engineering or Physics	
STAT 3502 [0.5]	Probability and Statistics	
MATH 3705 [0.5]	Mathematical Methods I	
9. 0.5 credit in:		0.5
COMP 1005 [0.5]	Introduction to Computer Science I	
10. 0.5 credit from:		0.5
MATH 3800 [0.5]	Mathematical Modeling and Computational Methods	
ECOR 2606 [0.5]	Numerical Methods	
11. 0.5 credit in:		0.5
ISAP 1000 [0.5]	Seminar in Science	
Approved courses outside the faculties of Science and Engineering and Design		
12. 1.5 credits in approved courses outside the faculties of Science and Engineering and Design		1.5
13. 1.0 credit in free electives.		1.0
Total Credits		20.0

Minor in Physics (4.0 credits)

The Minor in Physics is available to students registered in degree programs other than those offered by the Department of Physics. Careful attention must be paid to prerequisites.

Students are required to present a Minor CGPA of 4.00 or higher at graduation in order to be awarded a Minor in Physics.

Requirements

1. 1.0 credit from:	1.0
PHYS 1001 [0.5] Foundations of Physics I & PHYS 1002 [0.5] Foundations of Physics II	
PHYS 1003 [0.5] Introductory Mechanics and Thermodynamics & PHYS 1004 [0.5] Introductory Electromagnetism and Wave Motion	
PHYS 1007 [0.5] Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II (with an average grade of B- or higher)	
2. 2.0 credits in PHYS at the 2000-level	2.0
3. 1.0 credit in PHYS at the 3000- or 4000-level	1.0
Total Credits	4.0

B.Sc. Regulations

The regulations presented in this section apply to all Bachelor of Science programs. In addition to the requirements presented here, students must satisfy the University regulations common to all undergraduate students including the process of Academic Continuation Evaluation (see the *Academic Regulations of the University* section of this Calendar).

Breadth Requirement for the B.Sc.

Students in a Bachelor of Science program must present the following credits at graduation:

1. 2.0 credits in Science Continuation courses not in the major discipline; **students completing a double major are considered to have completed this requirement providing they have 2.0 credits in Science Continuation courses in each of the two majors;**
2. 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000)

In most cases, the requirements for individual B.Sc. programs, as stated in this Calendar, contain these requirements, explicitly or implicitly.

Students admitted to B.Sc. programs by transfer from another institution must present at graduation (whether taken at Carleton or elsewhere):

1. 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received fewer than 10.0 transfer credits; or,
2. 1.0 credit in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received 10.0 or more transfer credits.

Declared and Undeclared Students

Degree students are considered "Undeclared" if they have been admitted to a degree, but have not yet selected and been accepted into a program within that degree. The status "Undeclared" is available only in the B.A. and B.Sc. degrees. Undeclared students must apply to enter a program upon or before completing 3.5 credits.

Change of Program within the B.Sc. Degree

To transfer to a program within the B.Sc. degree, applicants must normally be *Eligible to Continue* (EC) in the new program, by meeting the CGPA thresholds described in Section 3.1.9 of the *Academic Regulations of the University*.

Applications to declare or change programs within the B.Sc. degree must be made online through Carleton Central by completing a Change of Program Elements (COPE) application form within the published deadlines. Acceptance into a program, or into a program element or option, is subject to any enrolment limitations, and/or specific program, program element or option requirements as published in the relevant Calendar entry.

Minors, Concentrations, and Specializations

Students may add a Minor, Concentration, or Specialization by completing a Change of Program Elements (COPE) application form online through Carleton Central. Acceptance into a Minor, Concentration, or Specialization normally requires that the student be *Eligible to Continue* (EC) and is meeting the minimum CGPAs described in Section 3.1.9 of the *Academic Regulations of the University*, as well as being subject to any specific requirements of the intended Minor, Concentration, or Specialization as published in the relevant Calendar entry.

Experimental Science Requirement

Students in a B.Sc. degree program must present at graduation at least two full credits of Experimental Science chosen from two different departments or institutes from the list below:

Approved Experimental Science Courses

Biochemistry	
BIOC 2200 [0.5]	Cellular Biochemistry
BIOC 3103 [0.5]	Experimental Biochemistry I: Principles and Practices
BIOC 3104 [0.5]	Experimental Biochemistry II: Research Experience
BIOC 4001 [0.5]	Methods in Biochemistry
BIOC 4201 [0.5]	Advanced Cell Culture and Tissue Engineering
Biology	
BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
BIOL 2001 [0.5]	Animals: Form and Function
BIOL 2002 [0.5]	Plants: Form and Function
BIOL 2104 [0.5]	Introductory Genetics
BIOL 2200 [0.5]	Cellular Biochemistry
BIOL 2600 [0.5]	Ecology
Chemistry	
CHEM 1001 [0.5]	General Chemistry I
CHEM 1002 [0.5]	General Chemistry II
CHEM 2103 [0.5]	Physical Chemistry I
CHEM 2203 [0.5]	Organic Chemistry I
CHEM 2204 [0.5]	Organic Chemistry II
CHEM 2302 [0.5]	Analytical Chemistry I
CHEM 2303 [0.5]	Analytical Chemistry II
CHEM 2800 [0.5]	Foundations for Environmental Chemistry
Earth Sciences	
ERTH 1002 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years
ERTH 2102 [0.5]	Mineralogy to Petrology
ERTH 2404 [0.5]	Engineering Geoscience
ERTH 2802 [0.5]	Field Geology I
ERTH 3111 [0.5]	Vertebrate Evolution: Mammals, Reptiles, and Birds
ERTH 3112 [0.5]	Vertebrate Evolution: Fish and Amphibians
ERTH 3204 [0.5]	Mineral Deposits
ERTH 3205 [0.5]	Physical Hydrogeology
Food Sciences	
FOOD 3001 [0.5]	Food Chemistry
FOOD 3002 [0.5]	Food Analysis
FOOD 3005 [0.5]	Food Microbiology
Geography	
GEOG 1010 [0.5]	Global Environmental Systems
GEOG 3108 [0.5]	Soil Properties
Neuroscience	
NEUR 3206 [0.5]	Sensory and Motor Neuroscience
NEUR 3207 [0.5]	Systems Neuroscience
NEUR 4600 [0.5]	Advanced Lab in Neuroanatomy
Physics	
PHYS 1001 [0.5]	Foundations of Physics I

PHYS 1002 [0.5]	Foundations of Physics II
PHYS 1003 [0.5]	Introductory Mechanics and Thermodynamics
PHYS 1004 [0.5]	Introductory Electromagnetism and Wave Motion
PHYS 1007 [0.5]	Elementary University Physics I
PHYS 1008 [0.5]	Elementary University Physics II
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3606 [0.5]	Modern Physics II
PHYS 3608 [0.5]	Modern Applied Physics

Course Categories for B.Sc. Programs

Science Geography Courses

GEOG 1010 [0.5]	Global Environmental Systems
GEOG 2006 [0.5]	Introduction to Quantitative Research
GEOG 2013 [0.5]	Weather and Water
GEOG 2014 [0.5]	The Earth's Surface
GEOG 3003 [0.5]	Quantitative Geography
GEOG 3010 [0.5]	Field Methods in Physical Geography
GEOG 3102 [0.5]	Geomorphology
GEOG 3103 [0.5]	Watershed Hydrology
GEOG 3104 [0.5]	Principles of Biogeography
GEOG 3105 [0.5]	Climate and Atmospheric Change
GEOG 3106 [0.5]	Aquatic Science and Management
GEOG 3108 [0.5]	Soil Properties
GEOG 4000 [0.5]	Field Studies
GEOG 4005 [0.5]	Directed Studies in Geography
GEOG 4013 [0.5]	Cold Region Hydrology
GEOG 4017 [0.5]	Global Biogeochemical Cycles
GEOG 4101 [0.5]	Two Million Years of Environmental Change
GEOG 4103 [0.5]	Water Resources Engineering
GEOG 4104 [0.5]	Microclimatology
GEOG 4108 [0.5]	Permafrost

Science Psychology Courses

PSYC 2001 [0.5]	Introduction to Research Methods in Psychology
PSYC 2002 [0.5]	Introduction to Statistics in Psychology
PSYC 2700 [0.5]	Introduction to Cognitive Psychology
PSYC 3000 [1.0]	Design and Analysis in Psychological Research
PSYC 3506 [0.5]	Cognitive Development
PSYC 3700 [1.0]	Cognition (Honours Seminar)
PSYC 3702 [0.5]	Perception
PSYC 2307 [0.5]	Human Neuropsychology I
PSYC 3307 [0.5]	Human Neuropsychology II

Science Continuation Courses

A course at the 2000 level or above may be used as a Science Continuation credit in a B.Sc. program if it is not in the student's major discipline, and is chosen from the following:

BIOC (Biochemistry)

BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biochemistry students may use BIOL 2005 only as a free elective.

CHEM (Chemistry)

COMP (Computer Science) A maximum of two half-credits at the 1000-level in COMP, excluding COMP 1001 may be used as Science Continuation credits.

ERTH (Earth Sciences), except ERTH 2415 which may be used only as a free elective for any B.Sc. program. Students in Earth Sciences programs may use ERTH 2401, ERTH 2402, and ERTH 2403 only as free electives.

Engineering. Students wishing to register in Engineering courses must obtain the permission of the Faculty of Engineering and Design.

ENSC (Environmental Science)

FOOD (Food Science and Nutrition)

GEOM (Geomatics)

HLTH (Health Sciences)

ISAP (Interdisciplinary Science Practice)

MATH (Mathematics)

NEUR (Neuroscience)

PHYS (Physics), except PHYS 2903

Science Geography Courses (see list above)

Science Psychology Courses (see list above)

STAT (Statistics)

TSES (Technology, Society, Environment) except TSES 2305. Biology students may use these courses only as free electives. Integrated Science and Environmental Science students may include these courses in their programs but may not count them as part of the Science Sequence.

Science Faculty Electives

Science Faculty Electives are courses at the 1000-4000 level chosen from:

BIOC (Biochemistry) except BIOC 1900 which may be used only as a free elective for any B.Sc. program.

BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biology & Biochemistry students may use BIOL 1010 and BIOL 2005 only as free electives.

CHEM (Chemistry) except CHEM 1003, CHEM 1004 and CHEM 1007

COMP (Computer Science) except COMP 1001

ERTH (Earth Sciences) except ERTH 1004 and ERTH 2415. Earth Sciences students may use ERTH 2401, ERTH 2402 and ERTH 2403 only as free electives.

Engineering

ENSC (Environmental Science)

FOOD (Food Science and Nutrition)

GEOM (Geomatics)

HLTH (Health Science)

ISAP (Interdisciplinary Science Practice)

MATH (Mathematics)

NEUR (Neuroscience)

PHYS (Physics) except PHYS 1901, PHYS 1902, PHYS 1905, and PHYS 2903

Science Geography (see list above)

Science Psychology (see list above)

STAT (Statistics)

TSES (Technology, Society, Environment) Biology students may use these courses only as free electives.

Advanced Science Faculty Electives

Advanced Science Faculty Electives are courses at the 2000-4000 level chosen from the Science Faculty Electives list above.

Approved Courses Outside the Faculties of Science and Engineering and Design (may include ISAP 1000)

All courses offered by the Faculty of Arts and Social Sciences, the Faculty of Public and Global Affairs, and the Sprott School of Business are approved as Arts or Social Sciences courses EXCEPT FOR: All Science Geography courses (see list above), all Geomatics (GEOM) courses, all Science Psychology courses (see list above). ISAP 1000 may be used as an Approved Course Outside the Faculties of Science and Engineering and Design.

Free Electives

Any course is allowable as a Free Elective providing it is not prohibited (see below). Students are expected to comply with prerequisite requirements and enrolment restrictions for all courses as published in this Calendar.

Courses Allowable Only as Free Electives in any B.Sc. Program

BIOL 4810 [0.5]	Education Research in Undergraduate Science
CHEM 1003 [0.5]	The Chemistry of Food, Health and Drugs
CHEM 1004 [0.5]	Drugs and the Human Body
CHEM 1007 [0.5]	Chemistry of Art and Artifacts
ERTH 1004 [0.5]	Earth's Epic Tale: A Story Across Billions of Years
ERTH 2415 [0.5]	Natural Disasters
ISCI 1001 [0.5]	Introduction to the Environment
ISCI 2000 [0.5]	Natural Laws
ISCI 2002 [0.5]	Human Impacts on the Environment
PHYS 1901 [0.5]	Planetary Astronomy
PHYS 1902 [0.5]	From our Star to the Cosmos
PHYS 1905 [0.5]	Physics Behind Everyday Life
PHYS 2903 [0.5]	Physics Towards the Future

Prohibited Courses

The following courses are not acceptable for credit in any B.Sc. program:

COMP 1001 [0.5]	Introduction to Computational Thinking for Arts and Social Science Students
MATH 1009 [0.5]	Mathematics for Business
MATH 1119 [0.5]	Linear Algebra: with Applications to Business
MATH 1401 [0.5]	Elementary Mathematics for Economics I

MATH 1402 [0.5] Elementary Mathematics for Economics II

all 0000-level courses

Co-operative Education

For more information about how to apply for the Co-op program and how the Co-op program works please visit the Co-op website.

All students participating in the Co-op program are governed by the Undergraduate Co-operative Education Policy.

Undergraduate Co-operative Education Policy Admission Requirements

Students can apply to Co-op in one of two ways: directly from high school, or after beginning a degree program at Carleton.

If a student applies to a degree program with a Co-op option from high school, their university grades will be reviewed two terms to one year prior to their first work term to ensure they meet the academic requirements after their first or second year of study. The time at which the evaluation takes place depends on the program of study. Students will automatically receive an admission decision via their Carleton email account.

Students who did not request Co-op at the time they applied to Carleton can request Co-op after they begin their university studies. To view application instructions and deadlines, please visit carleton.ca/co-op.

To be admitted to Co-op, a student must successfully complete 5.0 or more credits that count towards their degree, meet the minimum CGPA requirement(s) for the student's Co-op option, and fulfil any specified course prerequisites. To see the unique admission and continuation requirements for each Co-op option, please refer to the specific degree programs listed in the Undergraduate Calendar.

Participation Requirements

Co-op Participation Agreement

All students must adhere to the policies found within the Co-op Participation Agreement.

COOP 1000

Once a student has been admitted to the Co-op Program, they will be given access to register in COOP 1000. This zero-credit online course must be completed at least two terms prior to the student's first work term.

Communication with the Co-op Office

Students must maintain contact with the Co-op Office during their job search and while on a work term. All email communication will be conducted via the students' Carleton email account.

Employment

Although every effort is made to ensure a sufficient number of job postings for all Co-op students, no guarantee of employment can be made. The Co-op job search process is competitive, and success is dependent upon factors such as current market conditions, academic

performance, skills, motivation, and level of commitment to the job search. It is the student's responsibility to apply for positions via the Co-op job board in addition to actively conducting a self-directed job search. Students who do not obtain a co-op work term are expected to continue with their academic studies. It should be noted that hiring priority for positions within the Federal Government of Canada is given to Canadian citizens.

Registration

- Students must be registered as full-time during all fall and winter study terms beginning the term in which they enroll in COOP 1000.
- Students will be registered in a Co-op Work Term course while at work. This course does not carry academic course credit, but is noted on academic transcripts.
- Students may register in a 0.5 credit during a work term, provided the course is offered during the evening or is offered asynchronously online.
- Students must have at least one term of full-time studies left to complete following their final co-op work term. Students cannot end their degree on a work term.

Work Term Assessment and Evaluation

Work Term Evaluation

Employers are responsible for submitting to Carleton University final performance evaluations for their Co-op students at the end of their work terms.

Work Term Assessment

In order to successfully complete the co-op work term, students must receive a Satisfactory (SAT) grade on their Co-op Work Term Report, which they must submit at the completion of each four-month work term.

Graduation with the Co-op Designation

In order to graduate with the Co-op Designation, students must satisfy all requirements of the degree program in addition to the successful completion of three or four work terms (the number is dependent upon the student's academic program). Students found in violation of the Co-op Participation Agreement may have the Co-op Designation withheld.

Note: Participation in the co-op option will add up to one additional year for a student to complete their degree program.

Voluntary Withdrawal from the Co-op Option

Students who are currently on a co-op work term or who have already committed to a co-op work term either verbally or in writing may not leave the position and/or withdraw from the co-op option until they have completed the work term and all related requirements.

Involuntary or Required Withdrawal from the Co-op Option

Students may be removed from the Co-op Program for any of the following reasons:

1. Failure to achieve a grade of SAT in COOP 1000;
2. Failure to attend all interviews for positions to which the student has applied;
3. Declining more than one job offer during the job search;
4. Reneging on a co-op position that the student has accepted either verbally or in writing;
5. Continuing a job search after accepting a co-op position;
6. Dismissal from a work term by the co-op employer;
7. Leaving a work term without approval from the Co-op Management Team;
8. Receipt of an unsatisfactory work term evaluation;
9. Receiving a grade of UNS on the work term report.

International Students

All international students are required to possess a Co-op Work Permit issued by Immigration, Refugees and Citizenship Canada before they can begin working. The Co-operative Education Office will provide students with a letter of support to accompany their Co-op Work Permit application. Students are advised to discuss the application process and application requirements with the International Student Services Office.

Co-op Fees

All participating Co-op students are required to pay Co-op fees. For full details, please see the Co-op website.

B.Sc. Honours Physics, Applied Physics: Co-op Admission and Continuation Requirements

- Maintain full-time status in each study term;
- Be eligible to work in Canada (for off-campus work);
- Have successfully completed COOP 1000 .

In addition to the following:

1. Registered as a full-time student in the B.Sc. Honours Physics or Applied Physics program;
2. Successfully completed 5.0 or more credits;
3. Obtained an Overall CGPA of at least 6.50 and a Major CGPA of at least 8.00. These CGPAs must be maintained throughout the duration of the degree.

B.Sc. Honours Physics and Applied Physics students must successfully complete three (3) work terms to obtain the Co-op Designation.

Co-op Work Term Course: PHYS 3999

Work/Study Pattern:

Year 1		Year 2		Year 3		Year 4		Year 5	
Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern
Fall	S	Fall	S	Fall	S	Fall	W	Fall	S
Winter	S	Winter	S	Winter	S	Winter	W	Winter	S
Summer		Summer	W	Summer	W	Summer	W		

Legend

S: Study

W: Work

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System.

Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as *recommended*, it is not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Degree

- Bachelor of Mathematics (B. Math.) (Honours)
- Bachelor of Mathematics (B.Math.)

Admission Requirements

B.Math Honours

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include Advanced Functions, and Calculus and Vectors.

The overall admission cut-off average and/or the prerequisite course average may be considerably higher than the stated minimum requirements for admission to the combined B.Math./M.Sc. in Mathematics or Statistics.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be Eligible to Continue in their year level, in addition to

meeting the CGPA thresholds described in Section 3.1.9 of the Academic Regulations of the University. Advanced standing will be granted only for those subjects assessed as being appropriate for the program and the stream selected.

B.Math

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include Advanced Functions, and Calculus and Vectors.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be *Eligible to Continue* (EC) in their year level. Advanced standing will be granted only for those subjects assessed as being appropriate for the program and the stream selected.

Co-op Option

Direct Admission to the First Year of the Co-op Option

Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the Bachelor of Mathematics Honours program;
3. be eligible to work in Canada (for off-campus work placements).

Meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market (and thus the availability of co-op placement) may limit enrolment in the co-op option.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System. Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow

the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as recommended, it is not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Degrees

- B.Sc. (Honours)
- B.Sc. (Major)
- B.Sc.

Admission Requirements

B. Sc. Honours

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. For most programs including Biochemistry, Bioinformatics, Biotechnology, Chemistry, Combined Honours in Biology and Physics, Chemistry and Physics, Computational Biochemistry, Food Science, Nanoscience, Neuroscience and Biology, Neuroscience and Mental Health, and Psychology, the six 4U or M courses must include Advanced Functions, and two of Biology, Chemistry, Earth and Space Sciences, or Physics. (Calculus and Vectors is strongly recommended).

Specific Honours Admission Requirements

For the Honours programs in Earth Sciences, Environmental Science, Geomatics, Integrated Science, and Physical Geography, Calculus and Vectors may be substituted for Advanced Functions.

For the Honours programs in Physics and Applied Physics, and for double Honours in Mathematics and Physics, Calculus and Vectors is required in addition to Advanced Functions and one of 4U Physics, Chemistry, Biology, or Earth and Space Sciences. For all programs in Physics, 4U Physics is strongly recommended.

For Honours in Psychology, a 4U course in English is recommended.

For Honours in Environmental Science, a 4U course in Biology and Chemistry is recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be *Eligible to Continue* in their year level, in addition to meeting the CGPA thresholds described in Section 3.1.9 of the Academic Regulations of the University. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

B.Sc. Major and B.Sc.

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include Advanced Functions and two of Calculus and Vectors, Biology, Chemistry, Earth and Space Science, or Physics (Calculus and Vectors is strongly recommended). For the B.Sc. Major in Physics, 4U Physics is strongly recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be *Eligible to Continue* (EC) in their year level. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

Co-op Option

Direct Admission to the First Year of the Co-op Option

Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the Bachelor of Science Honours program;
3. be eligible to work in Canada (for off-campus work placements).

Note that meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market may limit enrolment in the co-op option.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Physics (PHYS) Courses

PHYS 1001 [0.5 credit]

Foundations of Physics I

This calculus-based course on classical mechanics covers kinematics, dynamics, gravitation, and oscillatory motion. This is a specialist course for students intending to take further courses in physics.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1002, BIT 1203, PHYS 1003, PHYS 1007, PHYS 1107.

Prerequisite(s): Grade 12 Mathematics: Advanced Functions and Grade 12 Mathematics: Calculus and Vectors or equivalent, plus one of MATH 1004 or MATH 1052 (the MATH course may be taken concurrently); or permission of the Physics Department. Grade 12 Physics is strongly recommended.

Lectures three hours a week, laboratory or tutorial three hours a week.

PHYS 1002 [0.5 credit]

Foundations of Physics II

An introduction to electricity, magnetism, electromagnetic fields, and wave motion. This is a specialist course for students intending to take further courses in physics.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1003 (no longer offered), BIT 1007, BIT 1204, PHYS 1004, PHYS 1008, PHYS 1108.

Prerequisite(s): PHYS 1001, or PHYS 1003, or PHYS 1007 with a grade of B-; MATH 1004 or MATH 1052 (may be taken concurrently); or permission of the Department.

Lectures three hours a week, laboratory or tutorial three hours a week.

PHYS 1003 [0.5 credit]

Introductory Mechanics and Thermodynamics

Mechanics, gravitation, oscillations, and thermodynamics. The application of calculus to solve problems in these areas of physics is introduced. This course is intended for students in the physical sciences and engineering.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1002, BIT 1203, PHYS 1001, PHYS 1007, PHYS 1107.

Prerequisite(s): Grade 12 Physics or equivalent, plus Grade 12 Mathematics: Advanced Functions or equivalent, plus one of MATH 1004 or MATH 1052 (the MATH course may be taken concurrently). Note that Grade 12 Mathematics: Calculus and Vectors is strongly recommended.

Lectures three hours a week, laboratory or tutorial three hours a week.

PHYS 1004 [0.5 credit]

Introductory Electromagnetism and Wave Motion

This calculus-based course introduces potential energy, work, electricity, magnetism, oscillations and waves.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1003 (no longer offered), BIT 1007, BIT 1204, PHYS 1002, PHYS 1008, PHYS 1108.

Prerequisite(s): MATH 1004, (ECOR 1033 and ECOR 1034)(the ECOR courses may be taken concurrently) or PHYS 1001 or PHYS 1003 or PHYS 1007 (a grade of at least B- is required for PHYS 1007), or permission of the Department.

Lectures three hours a week, laboratory or tutorial three hours a week.

PHYS 1007 [0.5 credit]

Elementary University Physics I

Mechanics, properties of matter, thermodynamics.

Applications chosen in part from the life sciences. For students who lack the prerequisites for PHYS 1001 or PHYS 1003, or who do not intend to take upper-year courses in physics.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1002, BIT 1203, PHYS 1001, PHYS 1003, PHYS 1107.

Prerequisite(s): (i) Grade 12 Mathematics: Advanced Functions or equivalent, or MATH 0107 (may be taken concurrently); or (ii) Grade 12 Mathematics: Calculus and Vectors or equivalent, or MATH 1007 (may be taken concurrently); or (iii) permission of the Physics Department.

Lectures three hours a week, laboratory or tutorial three hours per week.

PHYS 1008 [0.5 credit]

Elementary University Physics II

Electricity and magnetism, DC and AC circuits, wave motion and light. Elements of modern physics. Applications chosen in part from the life sciences.

Includes: Experiential Learning Activity

Precludes additional credit for BIT 1003 (no longer offered), BIT 1007, BIT 1204, PHYS 1002, PHYS 1004, PHYS 1108.

Prerequisite(s): PHYS 1001 or PHYS 1003 or PHYS 1007.

Lectures three hours a week, laboratory or tutorial three hours per week.

PHYS 1107 [0.5 credit]**Introductory University Physics I**

Mechanics, properties of matter, thermodynamics.

Applications chosen in part from the life sciences. For students who lack the prerequisites for PHYS 1001 or PHYS 1003, or who do not intend to take upper-year courses in physics.

Precludes additional credit for BIT 1002, BIT 1203, PHYS 1001, PHYS 1003, PHYS 1007.

Prerequisite(s): (i) Grade 12 Mathematics: Advanced Functions or equivalent, or MATH 0107 (may be taken concurrently); or (ii) Grade 12 Mathematics: Calculus and Vectors or equivalent, or MATH 1007 (may be taken concurrently); or (iii) permission of the Physics Department.

Lectures three hours a week.

PHYS 1108 [0.5 credit]**Introductory University Physics II**

Electricity and magnetism, DC and AC circuits, wave motion and light. Elements of modern physics.

Applications chosen in part from the life sciences. Precludes additional credit for BIT 1003 (no longer offered), BIT 1007, BIT 1204, PHYS 1002, PHYS 1004, PHYS 1008.

Prerequisite(s): PHYS 1001 or PHYS 1003 or PHYS 1007 or PHYS 1107.

Lectures three hours a week.

PHYS 1901 [0.5 credit]**Planetary Astronomy**

Description of the known stellar, galactic and extra-galactic systems together with the instruments used to study them. Modern ideas concerning the structure, origin and evolution of our own planet. Formation of the Moon - Earth system. Study of the planets in our solar system.

Precludes additional credit for PHYS 2203.

Lectures two and one-half hours a week.

PHYS 1902 [0.5 credit]**From our Star to the Cosmos**

Starting with the Sun, the course studies its composition and source of power, then compares our Sun with the other stars in the galaxy and beyond. Modern ideas concerning the structure, origin and evolution of the universe, pulsars and supernovae are examined.

Precludes additional credit for PHYS 2203.

Lectures two and one-half hours a week.

PHYS 1905 [0.5 credit]**Physics Behind Everyday Life**

Examination of the physics behind everyday life. Topics may include transportation, sports, weather and climate, electricity, and sustainable energy. No science background is required.

Includes: Experiential Learning Activity

Online Course.

PHYS 2004 [0.5 credit]**Modern Physics for Engineers**

Introduction to aspects of modern physics relevant to engineering. Thermal radiation. Concepts of relativistic kinematics. Wave-particle duality. Photoelectric effect, Compton scattering. Bohr theory of the hydrogen atom. Optical and x-ray spectra, lasers. Nuclear physics and applications.

Precludes additional credit for PHYS 2604 and PHYS 2605.

Prerequisite(s): PHYS 1002 or PHYS 1004 or PHYS 1008 with a grade of B- or better, plus MATH 1004 and MATH 1104 or equivalent. Restricted to B.Eng. students not in the Engineering Physics program. Students in programs other than B.Eng. must obtain permission of the Department.

Lectures three hours a week.

PHYS 2007 [0.5 credit]**Second Year Physics Laboratory: Selected Experiments and Seminars**

Students complete a number of experiments selected from classical physics and geometric optics, modern physics, etc. Seminars on relevant experimental topics will be included. Includes: Experiential Learning Activity.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 2202.

Prerequisite(s): PHYS 1002, or PHYS 1004 (PHYS 1008 is also acceptable provided a minimum average grade of B- is presented).

Laboratory, six hours a week.

PHYS 2101 [0.5 credit]**Mechanics and Properties of Matter**

Equations of motion for a single particle. Harmonic oscillation. Noninertial reference frames. Orbits in a central force field. Motion of systems of particles and of rigid bodies. Introduction to special relativity. Laboratory experiments in classical mechanics and properties of matter.

Includes: Experiential Learning Activity

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004, alternatively PHYS 1007 and PHYS 1008 with an overall average of B- or better or MATH 1004 and MATH 1104.

Lectures three hours a week, laboratory three hours a week, tutorials (optional) once a week.

PHYS 2202 [0.5 credit]**Wave Motion and Optics**

Geometrical optics. Types of waves, vibrating string and the classical wave equation. General solutions for traveling waves. Superposition and interference, coherence, wave packets, waves in 2 and 3 dimensions. Propagation of electromagnetic waves. Light and physical optics, oscillator model for dispersion, diffraction, polarization, and refraction.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 2212.

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004 (PHYS 1007 and PHYS 1008 are also acceptable provided a minimum average grade of B- is presented); plus MATH 1104 or MATH 2152, and MATH 2004 or MATH 2001 and MATH 2002 (MATH 2002 may be taken concurrently).

Lectures three hours a week, laboratory three hours a week.

PHYS 2203 [0.5 credit]**Astronomy**

The observational basis of astronomy. The history of astronomy, properties of light, solar system observations and stellar astronomy.

Precludes additional credit for PHYS 1901 and PHYS 1902.

Prerequisite(s): PHYS 1002 or PHYS 1004 or permission of the department. PHYS 1008 with a grade of B- or better may also be used if MATH 1004 or MATH 1007 or MATH 2052 have been successfully completed.

Lectures three hours a week.

PHYS 2212 [0.5 credit]**Wave Mechanics and Thermodynamics**

Types of waves and the classical wave equation, wave functions in 2 and 3 dimensions, reflection and refraction, superposition of waves, polarization, interference, diffraction, coherence, wave packets. Temperature and thermodynamic equilibrium, heat, work and first law of thermodynamics, entropy and second law of thermodynamics.

Precludes additional credit for PHYS 2202.

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004 (PHYS 1007 and PHYS 1008 are also acceptable provided a minimum average grade of B- is presented); plus MATH 1004 and MATH 1104 or MATH 2152, and MATH 2004 or MATH 2001 (MATH 2001 or MATH 2004 may be taken concurrently).

Lectures three hours a week

PHYS 2305 [0.5 credit]**Electricity and Magnetism**

Electrostatic field and potential, Gauss' law. Properties of conductors. Magnetic effects from currents. Motion of charges in electric and magnetic fields. Energy in electric and magnetic fields. Electromagnetic induction. Maxwell's equations in vacuum using vector differential and integral calculus.

Prerequisite(s): PHYS 1001, PHYS 1002, or PHYS 1003 and PHYS 1004, alternatively PHYS 1007 and PHYS 1008 with an overall grade of B- or higher; MATH 2004, or MATH 2001 and MATH 2002 (MATH 2002 may be taken concurrently).

Lectures three hours a week.

PHYS 2306 [0.5 credit]**Physics of Electrical and Electronic Measurements I**

D.C. and A.C. circuit theory. Resonant circuits. Basic measuring devices, the oscilloscope; impedances, bandwidth, noise; vacuum tubes, transistors, useful approximations for circuit design; feedback, amplifiers, oscillators; operational circuits; digital circuits. Lectures emphasize the physical basis of instrument design.

Laboratory emphasizes modern digital instrumentation.

Includes: Experiential Learning Activity

Prerequisite(s): PHYS 1001, PHYS 1002 or PHYS 1003 and PHYS 1004, alternatively PHYS 1007 and PHYS 1008 with an overall grade of B- or better.

Lectures three hours a week, laboratory three hours a week.

PHYS 2401 [0.5 credit]**Thermal Physics**

Introduction to thermodynamics and statistical mechanics. Temperature and thermodynamic equilibrium. Work, internal energy and heat; first law. Kinetic theory of gases. Basic probability theory. Microscopic states and entropy. Absolute temperature, reversibility and the second law of thermodynamics. Thermodynamic processes and applications.

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004, (PHYS 1007 and PHYS 1008 are also acceptable provided a minimum average grade of B-); plus MATH 1004 and MATH 1104 or MATH 2052 and MATH 2152.

Lectures three hours a week.

PHYS 2604 [0.5 credit]**Modern Physics I**

The course is designed to provide a logical transition from classical to modern physics. Special relativity. Rutherford scattering, atomic models. Thermal radiation. Photoelectric effect, Compton scattering. Bohr theory of the hydrogen atom. Atomic energy states, optical spectra, lasers. X-rays. Radioactivity. Quantum Mechanics.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 2004 and PHYS 2605.

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004 (PHYS 1007 and PHYS 1008 are also acceptable provided a minimum average grade of B- is presented); plus MATH 1004 and MATH 1104, or MATH 2052 and MATH 2152.

Lectures three hours a week, laboratory three hours a week.

PHYS 2605 [0.5 credit]**Modern Physics I**

The course is designed to provide a logical transition from classical to modern physics. Special relativity. Rutherford scattering, atomic models. Thermal radiation. Photoelectric effect, Compton scattering. Bohr theory of the hydrogen atom. Atomic energy states, optical spectra, lasers. X-rays. Radioactivity.

Precludes additional credit for PHYS 2004 and PHYS 2604.

Prerequisite(s): PHYS 1001 and PHYS 1002, or PHYS 1003 and PHYS 1004 (PHYS 1007 and PHYS 1008 are also acceptable provided a minimum average grade of B- is presented); plus MATH 1004 and MATH 1104, or MATH 2052 and MATH 2152.

Lecture 3 hours per week.

PHYS 2801 [0.5 credit]**Computational Methods in Physics**

Introduction to computational methods in physics. Software platforms and programming languages. Data formats and structures, histograms, and data visualization. Probability distributions, fitting/parameter estimation, function minimization. Interpretation and treatment of uncertainties. Introduction to machine learning and classification.

Prerequisite(s): PHYS 1001 or PHYS 1003 or PHYS 1007, and COMP 1005.

Lectures three hours a week.

PHYS 2903 [0.5 credit]**Physics Towards the Future**

From classical phenomena to aspects of modern physics and recent advances. Topics may include light and colour, music and sound, cell phones, the galaxy and beyond. No science background is required.

Includes: Experiential Learning Activity

Prerequisite(s): second-year standing.

Online course.

PHYS 3007 [0.5 credit]**Third Year Physics Laboratory: Selected Experiments and Seminars**

Students complete a small number of experiments selected from modern optics, holography, atomic physics, nuclear spectroscopy, radiation, etc. An exercise on literature searches and student seminars on experimental and numerical methods are included.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 3008, PHYS 3009.

Prerequisite(s): PHYS 2007 or (PHYS 2202 and PHYS 2604) or permission of the Department.

Laboratory, six hours a week.

PHYS 3008 [0.5 credit]**Third Year Physics Laboratory: Selected Experiments and Workshop**

Students complete a small number of experiments selected from modern optics, holography, atomic physics, nuclear spectroscopy, radiation, etc. Instruction on instrumentation building techniques will be given.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 3007, PHYS 3009.

Prerequisite(s): PHYS 2007 or (PHYS 2202 and PHYS 2604) or permission of the department.

Laboratory, six hours a week.

PHYS 3009 [0.5 credit]**Third Year Physics Laboratory: Selected Experiments and Seminars with Observational Astronomy**

Students complete a small number of experiments selected from astronomy, astrophysics, modern optics, holography, atomic physics, nuclear spectroscopy, radiation, etc. At least one astronomy/astrophysics related experiment is required. An exercise on literature searches and student seminars on experimental and numerical methods are included.

Includes: Experiential Learning Activity

Precludes additional credit for PHYS 3007, PHYS 3008.

Prerequisite(s): PHYS 2007 or (PHYS 2202 and PHYS 2604), and PHYS 2203, or permission of the Department.

Laboratory, six hours a week.

PHYS 3207 [0.5 credit]**Topics in Biophysics**

Introduction to biophysics. Random motion of molecules and diffusion; viscosity and the circulatory system; laws of thermodynamics and physical forces responsible for chemical reactions, molecular self-assembly and recognition; enzyme kinetics and molecular machines; nerve impulse and its propagation.

Prerequisite(s): PHYS 2212 or PHYS 2604, or permission of the Department.

Lectures three hours a week, tutorial or seminar one hour a week.

PHYS 3308 [0.5 credit]**Electromagnetism**

Electrostatics field and magnetostatics in the presence of matter. Solving Laplace's and Poisson's equations. Multipole expansions. Vector potential. Faraday's laws of induction; Maxwell's equations in matter. Waves in vacuum and dielectric media, guided waves.

Precludes additional credit for ELEC 3909.

Prerequisite(s): PHYS 2305 and MATH 3705, or permission of the Department.

Lectures three hours a week.

PHYS 3402 [0.5 credit]**Heat and Thermodynamics**

Zeroth, First, Second and Third Laws of Thermodynamics; enthalpy, Helmholtz and Gibbs functions and the Maxwell relations; phase transitions; thermodynamics of magnetism; cryogenics cooling by Joule-Thompson effect, adiabatic expansion of a gas, adiabatic demagnetization, helium dilution refrigeration; black body radiation; negative temperatures.

Prerequisite(s): PHYS 2101 and PHYS 2305, MATH 2007, MATH 2008, MATH 2107 and MATH 2401 or permission of the Department.

Lectures three hours a week.

PHYS 3606 [0.5 credit]**Modern Physics II**

Elements of condensed matter physics, semiconductors, superconductivity. Elements of nuclear physics, fission, fusion, power generation. Introduction to particle physics. Ionizing radiation: production, interactions, detection. Medical physics: radiation biophysics, cancer therapy, imaging.

Includes: Experiential Learning Activity

Also listed as PHYS 3608.

Prerequisite(s): PHYS 2007 and PHYS 2605, or PHYS 2604, and PHYS 3701, or permission of the Department.

Lectures three hours a week, laboratory two hours a week.

PHYS 3608 [0.5 credit]**Modern Applied Physics**

Elements of condensed matter physics, semiconductors, superconductivity. Modern optics. Elements of nuclear physics, fission, fusion, power generation. Ionizing radiation: production, interactions, detection. Medical physics: radiation biophysics, cancer therapy, imaging. Includes: Experiential Learning Activity

Also listed as PHYS 3606.

Prerequisite(s): PHYS 2007 and PHYS 2605, or PHYS 2604, and PHYS 3701, or permission of the Department.

Lectures three hours a week, laboratory three hours a week.

PHYS 3701 [0.5 credit]**Elements of Quantum Mechanics**

Analysis of interference experiments with waves and particles; fundamental concepts of quantum mechanics, Schrödinger equation; angular momentum, atomic beams; hydrogen atom; atomic and molecular spectroscopy; Pauli principle; simple applications in the physics of elementary particles.

Precludes additional credit for PHYS 3705.

Prerequisite(s): PHYS 2605 or PHYS 2604, MATH 2002 or MATH 2004 and MATH 3705 (may be taken concurrently), or permission of the Department.

Lectures three hours a week.

PHYS 3705 [0.5 credit]**Introduction to Quantum Systems**

This course is aimed at Computer Science and other students interested in developing a foundational understanding of quantum systems. Topics include: postulates of quantum mechanics, Hilbert space and observables, qubits, 2 state systems, entanglement, Schrodinger equation, 1D potentials, tunnelling, EPR paradox.

Precludes additional credit for PHYS 3701.

Prerequisite(s): (MATH 1004 or MATH 1007) and (MATH 1104 or MATH 1107).

Lectures three hours a week

PHYS 3801 [0.5 credit]**Classical Mechanics**

Introduction to Lagrangian and Hamiltonian mechanics: Poisson brackets, tensors and dyadics; rigid body rotations: introductory fluid mechanics coupled systems and normal coordinates; relativistic dynamics.

Prerequisite(s): PHYS 2101, PHYS 2202, PHYS 2305, MATH 2007, MATH 2008, MATH 2107, MATH 2401 or permission of the Department.

Lectures three hours a week.

PHYS 3802 [0.5 credit]**Advanced Dynamics**

Equations of motion for a single particle. Oscillatory Motion. Lagrangian and Hamiltonian formulations of mechanics. Central force motion. Motion of systems of particles and of rigid bodies.

Prerequisite(s): PHYS 2305, (MATH 2001 and MATH 2002) or MATH 2004, or permission of the Department.

Lectures three hours a week.

PHYS 3807 [0.5 credit]**Mathematical Physics I**

Boundary Value problems involving curvilinear coordinates; spherical harmonics, Bessel functions, Green's functions. Functions of a complex variable: analytic functions, contour integration, residue calculus.

Precludes additional credit for MATH 3007 or MATH 3057.

Prerequisite(s): ELEC 2501 or PHYS 2305, MATH 2004, MATH 3705 or permission of the Department.

Lectures three hours a week, tutorial one hour a week.

PHYS 3808 [0.5 credit]**Mathematical Physics II**

Solution of second-order total differential equations by Frobenius' method. Sturm-Liouville theory. Special functions: Legendre, Bessel, Hermite, Laguerre and associated functions. Partial differential equations: method of separation of variables, eigenfunctions and eigenvalues and eigenfunction expansions. Green's function techniques for solving inhomogeneous partial differential equations.

Precludes additional credit for MATH 3004, MATH 3008, MATH 3705, and PHYS 3806.

Prerequisite(s): PHYS 3807 or MATH 3007 or permission of the Department.

Lectures three hours a week.

PHYS 3999 [0.0 credit]**Co-operative Work Term Report**

Provides practical experience for students enrolled in the Co-operative option. Students must receive satisfactory evaluations from their work term employer. Written and oral reports will be required. Graded as Sat or Uns.

Includes: Experiential Learning Activity

Prerequisite(s): registration in the Physics Co-operative education option and permission of the Department.

PHYS 4007 [0.5 credit]**Fourth-Year Physics Laboratory: Selected Experiments and Seminars**

Students complete a small number of experiments selected from modern optics, holography, atomic physics, nuclear spectroscopy, radiation, etc. An exercise on literature searches and student seminars on experimental and numerical methods are included.

Includes: Experiential Learning Activity

Prerequisite(s): PHYS 3606 (or PHYS 3608) and registration in the Engineering Physics program.

Laboratory, six hours a week.

PHYS 4008 [0.5 credit]**Fourth-Year Physics Laboratory: Selected Experiments and Workshop**

Students complete a small number of experiments selected from modern optics, holography, atomic physics, nuclear spectroscopy, radiation, etc. Instruction on instrumentation building techniques will be given.

Includes: Experiential Learning Activity

Prerequisite(s): PHYS 3007.

Laboratory, six hours a week.

PHYS 4201 [0.5 credit]**Astrophysics**

Stellar evolution, including stellar modeling, main sequence stars, red giants and the end states of stars such as neutron stars and black holes. Galactic structure and dynamics. Neutrino astrophysics.

Prerequisite(s): PHYS 3701, PHYS 3606 or PHYS 3608, and PHYS 2401 or PHYS 4409, or permission of the Department. (PHYS 3606 or PHYS 3608 and PHYS 4409 may be taken concurrently).

Also offered at the graduate level, with different requirements, as PHYS 5401, for which additional credit is precluded.

Lectures three hours a week.

PHYS 4202 [0.5 credit]**Cosmology**

Observational evidence for the Big Bang. Cosmological space-time, expansion dynamics and contents of the universe. Physical processes in the expanding universe, inflation, nucleosynthesis, the cosmic microwave background, dark matter, and dark energy.

Prerequisite(s): PHYS 3701, PHYS 3606 or PHYS 3608, and PHYS 2401 or PHYS 4409, or permission of the Department. (PHYS 3606 or PHYS 3608 and PHYS 4409 may be taken concurrently).

Also offered at the graduate level, with different requirements, as PHYS 5402, for which additional credit is precluded.

Lectures three hours per week.

PHYS 4203 [0.5 credit]**Physical Applications of Fourier Analysis**

Fourier transform, convolution. Sampling theorem. Applications to imaging: descriptors of spatial resolution, filtering. Correlation, noise power. Discrete Fourier transform, FFT. Filtering of noisy signals. Image reconstruction in computed tomography and magnetic resonance. Laplace transform. Integral transforms, application to boundary value problems. Prerequisite(s): MATH 3705, or permission of the Department. Also offered at the graduate level, with different requirements, as PHYS 5313, for which additional credit is precluded. Lectures three hours a week.

PHYS 4208 [0.5 credit]**Modern Optics**

Electromagnetic wave propagation; reflection, refraction; Gaussian beams, guided waves. Laser theory: stimulated emission, cavity optics, modes, gain and bandwidth; atomic and molecular lasers. Mode locking, Q switching. Diffraction theory, coherence, Fourier optics, holography, laser applications. Optical communication systems, nonlinear effects: devices, fibre sensors, integrated optics. Prerequisite(s): PHYS 2212 or PHYS 2202, PHYS 3606 (or PHYS 3608), and PHYS 3308 or permission of the Department. Also offered at the graduate level, with different requirements, as PHYS 5318, for which additional credit is precluded. Lectures three hours a week.

PHYS 4307 [0.5 credit]**Electromagnetic Radiation**

Electromagnetic wave propagation in a vacuum, dielectrics, conductors, and ionized gases, reflection, refraction, polarization at the plane boundary between two media; waveguide and transmission line propagation; dipole and quadrupole radiation fields; antenna systems. Electromagnetic mass, radiation pressure. Tensor notation, transformation of the electromagnetic fields. Prerequisite(s): PHYS 3308, PHYS 3801, PHYS 3807 and PHYS 3808 (except for Mathematics and Physics Double Honours students), or permission of the Department. Lectures three hours a week.

PHYS 4407 [0.5 credit]**Statistical Physics**

Equilibrium statistical mechanics and its relation to thermodynamics. Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac statistics are derived, and applied in appropriate physical situations. Fluctuations. Kinetics and transport processes, including the Boltzmann transport equation and some of its applications. Prerequisite(s): PHYS 3402, PHYS 2602 or PHYS 3601, PHYS 3701 or PHYS 3602, PHYS 4707 (may be taken concurrently); or permission of the Department. Lectures three hours a week.

PHYS 4409 [0.5 credit]**Thermodynamics and Statistical Physics**

The three Laws of Thermodynamics, enthalpy, Helmholtz and Gibbs functions. Equilibrium statistical mechanics and its relation to thermodynamics. Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac statistics. Precludes additional credit for PHYS 3402 and PHYS 4407. Prerequisite(s): PHYS 3701 (may be taken concurrently), (MATH 2001 and MATH 2002) or MATH 2004, and MATH 3705, or permission of the Department.

PHYS 4508 [0.5 credit]**Solid State Physics**

An introduction to solid state physics. Topics include crystal structure, phonons and lattice vibrations, conductors, semiconductors, insulators and superconductivity. Prerequisite(s): PHYS 3606 or PHYS 3608, and PHYS 3701, or permission of the Department. Lectures three hours a week.

PHYS 4602 [0.5 credit]**Physics of Elementary Particles**

Standard Model. Properties of leptons, quarks, hadrons. Fundamental interactions: photon, gluons, W/Z bosons. Higgs boson. Conservation laws, invariance principles, quantum numbers. Decay rates and scattering cross-sections. Quantum electrodynamics and chromodynamics. Resonances. Weak interactions, CKM matrix, parity and CP violation. Neutrino masses and oscillations. Future directions. Prerequisite(s): PHYS 4707 or permission of the Department. Also offered at the graduate level, with different requirements, as PHYS 5602, for which additional credit is precluded. Lectures three hours a week.

PHYS 4608 [0.5 credit]**Nuclear Physics**

Ground state properties of nuclei. Nuclear models, binding energy, properties of excited nuclei. Alpha, beta and gamma decay. Passage of radiation through matter, detectors. Nuclear reactions, cross sections, fission, fusion. Elements of neutron physics.

Prerequisite(s): PHYS 3606 or PHYS 3608 or permission of the Department.

Lectures three hours a week.

PHYS 4705 [0.5 credit]**Introduction to Quantum Phenomena**

Postulates of Quantum Mechanics, Quantum communication and cryptography, Entanglement, EPR paradox and Bell inequalities, Introduction to wave mechanics: Energy quantization and tunneling, Survey of quantum material: superconductor, topological insulators, semimetals. Survey of quantum sensing: NMR, trapped ions, superconducting qubit, atomic clocks.

Also offered at the graduate level, with different requirements, as PHYS 5705, for which additional credit is precluded.

Lectures three hours per week.

PHYS 4707 [0.5 credit]**Introduction to Quantum Mechanics I**

The basic interpretative postulates of quantum mechanics; applications of wave mechanics and operator methods to various quantum mechanical systems; quantum mechanical treatment of angular momentum.

Prerequisite(s): PHYS 3701 and PHYS 3807 or equivalent, or permission of the Department.

Lectures three hours a week.

PHYS 4708 [0.5 credit]**Introduction to Quantum Mechanics II**

Scattering theory and application; bound state problems; approximation methods.

Prerequisite(s): PHYS 4707 or permission of the Department.

Lectures three hours a week.

PHYS 4804 [0.5 credit]**Introduction to General Relativity**

Special relativity using tensor analysis. Curved spacetime with physics applications which may include the solar system, stars, black holes and gravitational waves. Introduction to differential geometry and Einstein's field equations.

Prerequisite(s): PHYS 3802 or equivalent, or permission of the Department.

Also offered at the graduate level, with different requirements, as PHYS 5804, for which additional credit is precluded.

Lectures three hours a week.

PHYS 4807 [0.5 credit]**Statistical Data Analysis Techniques for Physics**

Computational methods used in analysis of experimental data. Introduction to probability and random variables. Monte Carlo methods for simulation of random processes. Statistical methods for parameter estimation and hypothesis tests. Confidence intervals. Multivariate data classification. Unfolding methods. Examples primarily from particle and medical physics.

Prerequisite(s): third year standing in a physics program and an ability to program in Python, Java, C or C++, and permission of the Department.

Also offered at the graduate level, with different requirements, as PHYS 5002, for which additional credit is precluded.

Lectures three hours a week.

PHYS 4901 [0.5 credit]**Special Topics in Physics**

Each year, at the direction of the Department, a course on a special topic may be offered.

Prerequisite(s): permission of the Department.

PHYS 4907 [0.5 credit]**Fourth-Year Project**

Advanced projects of an experimental or theoretical nature with an orientation towards research. A written mid-term progress report is required and also a written and oral report at the conclusion of the project.

Includes: Experiential Learning Activity

Prerequisite(s): fourth-year standing in an Honours Physics program or equivalent, and permission of the Department.

Project. Fall term only.

PHYS 4908 [0.5 credit]**Fourth-Year Project**

Advanced projects of an experimental or theoretical nature with an orientation towards research. A written mid-term progress report is required and also a written and oral report at the conclusion of the project.

Includes: Experiential Learning Activity

Prerequisite(s): fourth-year standing in an Honours Physics program or equivalent, and permission of the Department.

Project. Winter term only.

PHYS 4909 [1.0 credit]**Fourth-Year Project**

Advanced projects of an experimental or theoretical nature with an orientation towards research. A written mid-term progress report is required and also a written and oral report at the conclusion of the project.

Includes: Experiential Learning Activity

Prerequisite(s): fourth-year standing in an Honours Physics program or equivalent, and permission of the Department.
Project